

Zhifan Luo

Ph.D. Student · Department of Data and Systems Engineering · The University of Hong Kong
luozf0105@gmail.com · sio-2.github.io · github.com/SiO-2 · [ORCID](#)

Education

Ph.D. in Data and Systems Engineering <i>The University of Hong Kong</i> • Advisor: Prof. Shinan Liu	Sep. 2026 – Present <i>Hong Kong SAR</i>
M.S. in Cyberspace Security <i>Zhejiang University</i> • Advisor: Prof. Zhan Qin	Sep. 2023 – Jun. 2026 <i>Hangzhou, China</i>
B.S. in Information Security <i>Zhejiang University</i>	Sep. 2019 – Jun. 2023 <i>Hangzhou, China</i>

Research Interests

LLM security and privacy; machine learning systems; secure and efficient LLM inference; KV-cache privacy and side-channel analysis.

Publication

Zhifan Luo, Shuo Shao, Su Zhang, Lijing Zhou, Yuke Hu, Chenxu Zhao, Zhihao Liu, and Zhan Qin. “Shadow in the Cache: Unveiling and Mitigating Privacy Risks of KV-cache in LLM Inference.” *Network and Distributed System Security Symposium (NDSS)*, 2026. arXiv · Code

Research Experience

Doctoral Research <i>The University of Hong Kong</i> • Researching security and privacy in modern LLM inference systems under the supervision of Prof. Shinan Liu. • Current interests include KV-cache protection, inference side channels, and practical systems integration.	Sep. 2026 – Present <i>Hong Kong SAR</i>
Graduate Research Assistant <i>State Key Laboratory of Blockchain and Data Security, Zhejiang University</i> • Studied privacy leakage from KV-caches and designed KV-Cloak, a lightweight defense for LLM inference. • Led the resulting work published at NDSS 2026.	Sep. 2023 – Jun. 2026 <i>Hangzhou, China</i>

Technical Skills

Programming	Python, C/C++
ML & Systems	PyTorch, Transformers, vLLM
Tools	Git, Docker, LaTeX

Last updated September 2026